

S.No	Author & Year	Methodology	Performance	Advantages	Limitations
1	Deep Learning-Based Automatic Detection of Tuberculosis Disease in Chest X-Ray Images (2022)	Utilized transfer learning models including VGG16, VGG19, ResNet50, Xception, and InceptionV3 for automated TB detection from chest X-ray images.	Achieved approximately 90% classification accuracy.	Demonstrated effectiveness of transfer learning for TB diagnosis. Reduced training time compared to training CNNs from scratch.	Limited to binary classification. No explainable AI mechanism. Small dataset size affected generalization.
2	Deep and Hybrid Learning Technique Using Feature Fusion for Early TB Detection (2022)	Combined handcrafted features (GLCM, LBP, DWT) with deep features extracted from ResNet50 and GoogLeNet. Feature fusion was performed before classification.	Accuracy exceeded 99%.	Improved feature representation by combining deep and handcrafted features. Enhanced classification performance.	Increased computational complexity. Lack of explainability and localization mechanisms.
3	Ensemble Technique Coupled with Deep Transfer Learning for Tuberculosis Detection (2022)	Applied U-Net based lung segmentation followed by ensemble CNN classification models.	Accuracy around 98%.	Segmentation improved focus on lung regions. Ensemble learning enhanced robustness.	Explainability techniques were not incorporated. Computationally expensive.
4	CBAMWDNet: Improved DenseNet with Attention Mechanism for TB Detection (2023)	DenseNet architecture enhanced using Convolutional Block Attention Module (CBAM) to improve feature extraction and attention learning.	Accuracy around 98.8%.	Attention modules improved feature learning. Better localization compared to traditional CNNs.	Focused only on classification. No Explainable AI framework included.
5	Deep Learning Architecture for Multi-Class Lung Disease Classification (2023)	CNN-based framework designed for classification of multiple lung diseases including TB, pneumonia, COVID-19, and normal cases.	Accuracy around 96%.	Demonstrated capability for multi-class disease classification.	Limited interpretability. No localization or explanation mechanism provided.

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6	Machine and Deep Learning for Tuberculosis Detection on Chest X-Rays: Systematic Literature Review (2023)	Comprehensive review of ML and DL techniques used for TB diagnosis. Analyzed CNNs, SVMs, Random Forests, and deep learning models.	Reported pooled sensitivity and specificity above 98%.	Identified strengths and weaknesses of current approaches. Highlighted the growing importance of explainable AI.	Most existing systems lacked deployment-ready explainability and clinical validation.
7	TB-CXRNet: Tuberculosis and Drug-Resistant Tuberculosis Detection Technique Using Chest X-Ray Images (2024)	Deep CNN architecture for TB, Non-TB, Healthy, MDR-TB, and XDR-TB classification. Integrated Score-CAM for interpretability.	Accuracy of 93.32% for TB classification.	Large-scale dataset. Included drug-resistant TB classification and explainability.	Relied mainly on CNN architecture. Did not utilize transformer-based feature learning.
8	Tuberculosis Detection from Chest X-Ray Image Modalities Based on Transformer and CNN (2024)	Hybrid architecture combining Data Efficient Image Transformer (DeiT) and ResNet16. Transformer captured global features while CNN extracted local patterns.	Accuracy 99.38%, Sensitivity 99.49%, Specificity 99.26%.	Combined strengths of CNNs and Transformers. Lightweight architecture with fast inference speed.	No lung segmentation stage. Limited explainability and clinical reporting.
9	Enhanced CoAtNet Based Hybrid Deep Learning Architecture for Automated Tuberculosis Detection in Human Chest X-Rays (2025)	Utilized CoAtNet architecture combining CNN and Vision Transformer components. Applied CLAHE, augmentation, oversampling, and LIME explainability.	Accuracy 86.39%, ROC-AUC 93.79%.	Hybrid architecture captured both local and global features. Included explainable AI through LIME.	No lung segmentation. Relied on LIME instead of Grad-CAM. No automatic report generation.
10	Ensemble Deep Learning Architectures for Tuberculosis Detection (2026)	Ensemble framework using multiple deep learning models with multi-scale feature extraction and feature fusion.	AUC approximately 0.98.	Improved robustness and generalization performance. Reduced model bias.	High computational requirements. Limited interpretability and transparency.